

Unit 1, Lesson 8: Equivalent Radical Expressions



Benchmark

MA.912.NSO.1.4 Apply previous understanding of operations with rational numbers to add, subtract, multiply, and divide numerical radicals.



Learning Targets

- ☐ I can relate the power of a product law of exponents to the product law of radicals.
- ☐ Given a numerical radical, I can use the product law of radicals to write an equivalent numerical radical.



1.8.1 Warm-Up: Error Analysis

Leilana and Khamari wrote different numerical expressions and expanded them to show each has a value of 16.

Leilana's Work	Khamari's Work
$2^4 = 2 \cdot 2 \cdot 2 \cdot 2 = 16$	$2^8 = 2 \cdot 8 = 16$

1. Explain whether you agree with Leilana's work or Khamari's work.
2. Write another numerical expression involving exponents with a value of 16.



1.8.2 Exploration: Equivalent Radical Expressions

Ameer was asked to write an equivalent expression for $\sqrt{18}$. His work and resulting expression are shown.

Given Expression	Ameer's Work	Equivalent Expression
$\sqrt{18}$	$18^{\frac{1}{2}} = (6 \cdot 3)^{\frac{1}{2}} = (6)^{\frac{1}{2}} \cdot (3)^{\frac{1}{2}}$	$\sqrt{6} \cdot \sqrt{3}$

Sophia wrote a different equivalent expression for $\sqrt{18}$. Her work and resulting expression are shown.

Given Expression	Sophia's Work	Equivalent Expression
$\sqrt{18}$	$18^{\frac{1}{2}} = (9 \cdot 2)^{\frac{1}{2}} = 9^{\frac{1}{2}} \cdot 2^{\frac{1}{2}} = \sqrt{9} \cdot \sqrt{2}$	$3\sqrt{2}$

- What similarities and differences do you notice between Ameer's work and Sophia's work?

Similarities	Differences

- With your partner, complete the work to find equivalent expressions for $\sqrt{175}$ two different ways.

Given Expression	Work	Equivalent Radical Expression
$\sqrt{175}$	$175^{\frac{1}{2}} = (35 \cdot 5)^{\frac{1}{2}} = (\quad)^{\frac{1}{2}} \cdot (\quad)^{\frac{1}{2}}$	$\sqrt{\quad} \cdot \sqrt{\quad}$
$\sqrt{175}$	$175^{\frac{1}{2}} = (25 \cdot 7)^{\frac{1}{2}} = (\quad)^{\frac{1}{2}} \cdot (\quad)^{\frac{1}{2}}$	$\sqrt{\quad} \cdot \sqrt{\quad}$

- Which equivalent radical expression above can be rewritten in the form of an **integer** multiplied by a **radical**?

$\sqrt{\quad} \cdot \sqrt{\quad} = \quad \sqrt{\quad}$ because $\quad$ is a perfect square, therefore $\sqrt{\quad} = \quad$.



1.8.3 Guided Instruction: Equivalent Radical Expressions

Recall the **power of a product** law of exponents. It states the product of two or more numbers raised to a power is equal to the product of each number raised to the same power.

$$(ab)^m = a^m \cdot b^m \text{ or } a^m \cdot b^m = (ab)^m$$

The **product law of radicals** works similarly. Each of these equivalent expressions demonstrates the product law of radicals.

$$\begin{aligned}\sqrt{18} &= \sqrt{6} \cdot \sqrt{3} \\ \sqrt{18} &= \sqrt{9} \cdot \sqrt{2}\end{aligned}$$

$$\begin{aligned}\sqrt{175} &= \sqrt{35} \cdot \sqrt{5} \\ \sqrt{175} &= \sqrt{25} \cdot \sqrt{7}\end{aligned}$$

When a radical expression is rewritten as a product of its factors, each of the radicals will have the same index.

Product Law of Radicals

$$\sqrt[m]{ab} = \sqrt[m]{a} \cdot \sqrt[m]{b} \text{ or } \sqrt[m]{a} \cdot \sqrt[m]{b} = \sqrt[m]{ab}$$

1. Rewrite $\sqrt[3]{325}$ in two different ways using the product law of radicals.

When rewriting radical expressions, mathematicians generally try to rewrite the expression in the form of an **integer** multiplied by a **radical**, when possible.

$$\sqrt{18} = \sqrt{9} \cdot \sqrt{2}, \text{ which can be rewritten as } 3\sqrt{2}$$

$$\sqrt{175} = \sqrt{25} \cdot \sqrt{7}, \text{ which can be rewritten as } 5\sqrt{7}$$

2. What was done to rewrite the example expressions?

3. The following radical expressions are equivalent to $\sqrt[3]{320}$.

$$\sqrt[3]{32} \cdot \sqrt[3]{10}$$

$$\sqrt[3]{64} \cdot \sqrt[3]{5}$$

$$\sqrt[3]{8} \cdot \sqrt[3]{40}$$

$$\sqrt[3]{16} \cdot \sqrt[3]{20}$$

- a. Determine which of the expressions can be rewritten in the form of an integer multiplied by a radical.
- b. Discuss with your partner why these expressions can be rewritten.

4. Some radical expressions are given as an integer multiplied by the radical before any rewriting is done. Priya and Raj were asked to rewrite this type of expression. Their work is shown.

Priya's Work	$3\sqrt{20} = 3 \cdot \sqrt{4} \cdot \sqrt{5} = 3 \cdot 2\sqrt{5} = 6\sqrt{5}$
Raj's Work	$3\sqrt{20} = 3 \cdot \sqrt{4} \cdot \sqrt{5} = 3 \cdot 2\sqrt{5} = 5\sqrt{5}$

Explain whether Priya or Raj rewrote the expression correctly.



1.8.4 Try It: Equivalent Radical Expressions

1. Write two equivalent expressions for each radical expression. Show all your steps.

a.

$\sqrt{135}$
$\sqrt{135} =$
$\sqrt{135} =$

b.

$\sqrt[3]{600}$
$\sqrt[3]{600} =$
$\sqrt[3]{600} =$

2. Write an equivalent expression for the following radical expression.

$$5\sqrt[3]{56}$$



1.8.5 Your Turn!

1. Select all expressions that are equivalent to $\sqrt{40}$.

- ☐ $(40)^{\frac{1}{2}}$
- ☐ $2\sqrt{20}$
- ☐ $\sqrt{4} \cdot \sqrt{10}$
- ☐ $(20 \cdot 2)^{\frac{1}{2}}$
- ☐ $5\sqrt{8}$
- ☐ $2\sqrt{10}$
- ☐ $(10 \cdot 4)^{\frac{1}{2}}$

2. Write an equivalent expression for $\sqrt[3]{378}$ in the form of a constant multiplied by a radical.



1.8.6 Wrap-Up: Reflection

1. What was the **muddiest** (most difficult or confusing) point in today's lesson on writing equivalent expressions for numerical radicals?

